

Lab_09

Kenya Sanchez

Section 1: Background

Learning Objectives

Section 2: Importing Candy Data

Download Data from FiveThirtyEight GitHub

```
# Download data using url
url <- "https://raw.githubusercontent.com/fivethirtyeight/data/master/candy-power-ranking/candy_data.csv"

# Create a vector the the data
candy_file <- "candy-data.csv"
candy = read.csv(url, row.names=1)
head(candy)
```

	chocolate	fruity	caramel	peanut	almond	nougat	crisped	rice	wafer
100 Grand	1	0	1		0	0			1
3 Musketeers	1	0	0		0	1			0
One dime	0	0	0		0	0			0
One quarter	0	0	0		0	0			0
Air Heads	0	1	0		0	0			0
Almond Joy	1	0	0		1	0			0

	hard	bar	pluribus	sugar	percent	price	percent	win	percent
100 Grand	0	1	0	0.732	0.860	66.97173			
3 Musketeers	0	1	0	0.604	0.511	67.60294			
One dime	0	0	0	0.011	0.116	32.26109			
One quarter	0	0	0	0.011	0.511	46.11650			
Air Heads	0	0	0	0.906	0.511	52.34146			
Almond Joy	0	1	0	0.465	0.767	50.34755			

2.1: What is in the Data Set?

Q1: How many different candy types are in this dataset?

Answer Q1: There are **85 different candy types** in this data set. There are also 12 different attributions and variables used to characterize each candy types

```
# Candy Types and Attributions  
dim(candy)
```

```
[1] 85 12
```

```
# Candy Types ONLY  
nrow(candy)
```

```
[1] 85
```

Q2: How many fruity candy types are in the dataset

Answer Q2: There are **38 fruity candy types** in this data set. The table function() allows us to see how many candy types are characterized as fruity and how many are not.

```
# Types of Candy corresponding to Fruity Characteristic  
# Using "==" Labels True and False rather than 1 and 0  
table(candy$fruity==1)
```

```
FALSE  TRUE  
   47    38
```

2.2: What is your favorite candy?

winpercent(): Percentage of People that prefer *this* candy over another randomly chosen candy

Twix Example: 81.64% of people prefer twix candy over any other randomly chosen candy from the data set.

```
# Call out winpercent using R
candy["Twix", ]$winpercent
```

```
[1] 81.64291
```

Alternative Method:

```
# Call out winpercent using dplyr package
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
candy |>
  filter(row.names(candy)=="Twix") |>
  select(winpercent)
```

```
      winpercent
Twix    81.64291
```

Q3: What is your favorite candy (other than Twix) in the dataset and what is its winpercent value?

Answer Q3: My favorite candy from this data set is “**Snickers**”, its winpercent value is **76.67%**.

```
# Winpercent of Snickers Candy
candy |>
  filter(row.names(candy)=="Snickers") |>
  select(winpercent)
```

```

      winpercent
Snickers    76.67378

```

Q4: What is the winpercent value for “Kit Kat”?

Answer Q4: The winpercent value for “Kit Kat” is: **76.77%**.

```

candy |>
  filter(row.names(candy)=="Kit Kat") |>
  select(winpercent)

```

```

      winpercent
Kit Kat    76.7686

```

Q5 What is the winpercent value for “Tootsie Roll Snack Bars”?

Answer Q5: The winpercent value for “Tootsie Roll Snack Bars” is: **49.65%**.

```

candy |>
  filter(row.names(candy)=="Tootsie Roll Snack Bars") |>
  select(winpercent)

```

```

      winpercent
Tootsie Roll Snack Bars    49.6535

```

Downloading and Trying Skimr package

```

library("skimr")
skim(candy)

```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyal- mondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedrice- wafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6: Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

Answer Q6: Yes, the columns for the winpercent observation appear to be on a different scale compared to the majority / the rest of the data set. The winpercent columns appear to be on a 0-100 % scale whereas every other column is displayed on a 0-1 scale thus giving us decimals rather than percentages.

Q7: What do you think a zero and one represent for the candy\$chocolate column?

Answer Q7: For the candy\$chocolate column, 0 and 1 are used as False and True markers (respectively). Having a 1 means that there IS chocolate in that candy.

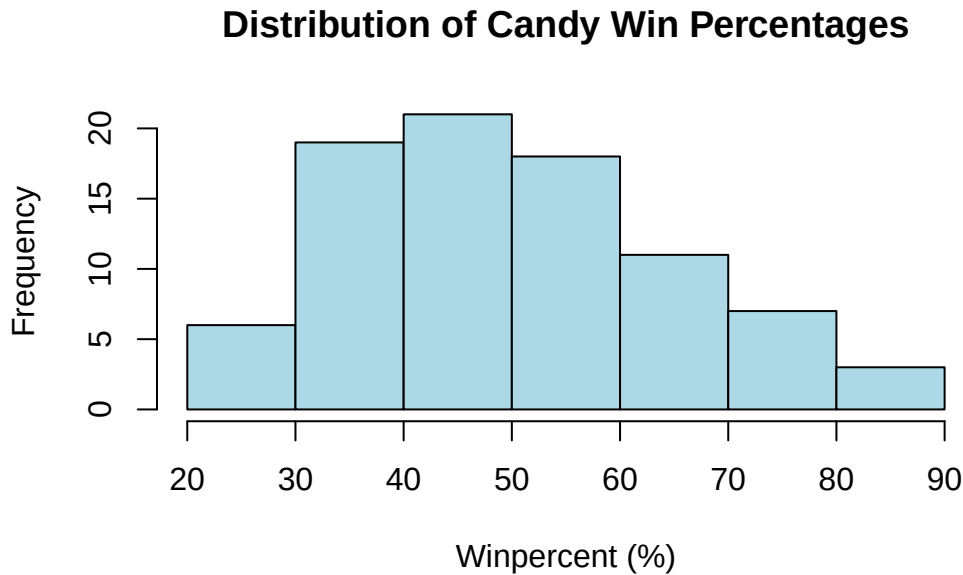
Section 3: Exploratory Analysis

Q8: Plot a histogram of winpercent values using both base R and ggplot2.

Answer Q8: Below are the histograms of winpercent using both R and ggplot2

```
# Histogram using R
```

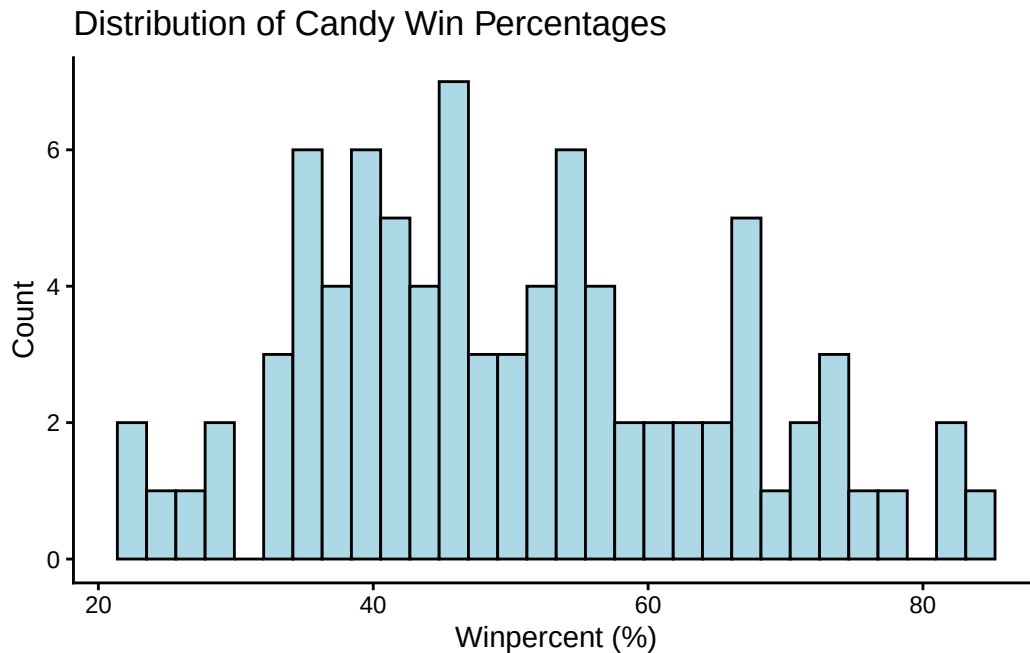
```
hist(candy$winpercent,  
     main = "Distribution of Candy Win Percentages",  
     xlab = "Winpercent (%)",  
     col = "lightblue",  
     border = "black")
```



```
# Histogram using ggplot
```

```
library(ggplot2)  
  
ggplot(candy) +  
  aes(winpercent) +  
  geom_histogram( fill = "lightblue", color = "black") +  
  scale_x_continuous(breaks = seq(0,100, by = 20)) +  
  labs(title = "Distribution of Candy Win Percentages",  
       x = "Winpercent (%)",  
       y = "Count") +  
  theme_classic()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



Q9: Is the distribution of winpercent values symmetrical?

Answer Q9: No, the overall distribution of winpercent is NOT symmetrical. There is a slight skew towards the right of both histograms.

Q10: Is the center of the distribution above or below 50%?

Answer Q10: The center of distribution is below 50%. This means that the majority of the Candies have a winpercent that is less than 50%.

Q11: On average is chocolate candy higher or lower ranked than fruit candy?

Answer Q11: Yes, on average chocolate candy is ranked higher than fruit candy. We know this because the winpercent of chocolate is 60.92% and the winpercent of fruit candy is 44.12%. On average chocolate is ranked higher than fruit candy by about 16%.

```
# Mean winpercent for chocolate
mean(candy$winpercent[candy$chocolate == 1])
```

```
[1] 60.92153
```

```
# Mean winpercent for fruity  
mean(candy$winpercent[candy$fruity == 1])
```

```
[1] 44.11974
```

Q12: Is this difference statistically significant?

Answer Q12: Yes, this difference is statistically significant. After running the t-test the p value results in 2.871e-08 which is much lower than the significance threshold of 0.05. Since the p value is smaller than 0.05, we conclude that the difference between chocolate and fruity candy is highly significant. It's not just a coincidence in this specific survey. (T-test shown below)

```
# Define the two groups using the logical trick  
chocolate <- candy$winpercent[as.logical(candy$chocolate)]  
fruity     <- candy$winpercent[as.logical(candy$fruity)]  
  
# Run the t-test  
t.test(chocolate, fruity)
```

Welch Two Sample t-test

```
data: chocolate and fruity  
t = 6.2582, df = 68.882, p-value = 2.871e-08  
alternative hypothesis: true difference in means is not equal to 0  
95 percent confidence interval:  
 11.44563 22.15795  
sample estimates:  
mean of x mean of y  
 60.92153  44.11974
```

Section 4: Overall Candy Rankings

Q13: What are the five least liked candy types in this set?

Answer Q:13 The five least liked candy types in this set are: “Nik L Nip”, “Boston Baked Beans”, “Chiclets”, “Super Bubble”, and “Jawbusters”. This is shown in the ascending winpercent table below.


```
# Sort by winpercent (ascending) and take the top 5
head(candy[order(candy$winpercent), ], 5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Nik L Nip	0	1	0		0	0
Boston Baked Beans	0	0	0		1	0
Chiclets	0	1	0		0	0
Super Bubble	0	1	0		0	0
Jawbusters	0	1	0		0	0

	crisped	rice	wafer	hard	bar	pluribus	sugar	percent	price	percent
Nik L Nip				0	0	0	1	0.197		0.976
Boston Baked Beans				0	0	0	1	0.313		0.511
Chiclets				0	0	0	1	0.046		0.325
Super Bubble				0	0	0	0	0.162		0.116
Jawbusters				0	1	0	1	0.093		0.511

	winpercent
Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744

Q14: What are the top 5 all time favorite candy types out of this set?

Answer Q:14 The top 5 all time favorite candy types out of this set are “Reese’s Peanut Butter cup”, “Reese’s Miniatures”, “Twix”, “Kit Kat”, and “Snickers”. As shown below, these results were botained by listing winpercent of candy dataset in descreasing order (from highest to lowerst win percents).

```
# Sort by winpercent in descending order and take the top 5
head(candy[order(candy$winpercent, decreasing = TRUE), ], 5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Reese's Peanut Butter cup	1	0	0		1	0
Reese's Miniatures	1	0	0		1	0
Twix	1	0	1		0	0
Kit Kat	1	0	0		0	0
Snickers	1	0	1		1	1

	crisped	rice	wafer	hard	bar	pluribus	sugar	percent
Reese's Peanut Butter cup				0	0	0		0.720
Reese's Miniatures				0	0	0		0.034

Twix	1	0	1	0	0.546
Kit Kat	1	0	1	0	0.313
Snickers	0	0	1	0	0.546

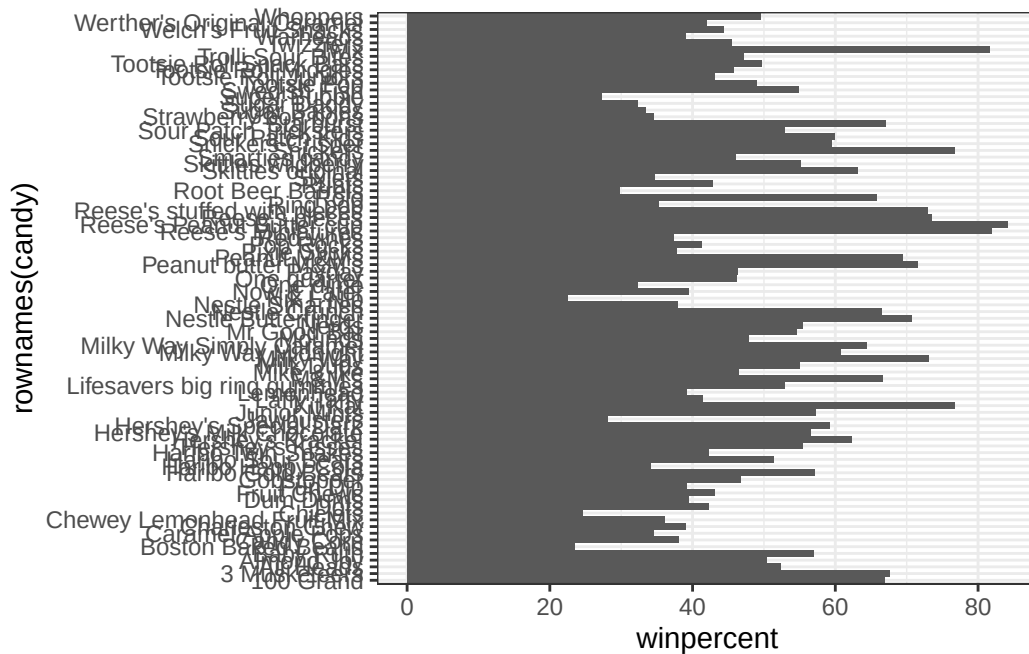
	pricepercent	winpercent
Reese's Peanut Butter cup	0.651	84.18029
Reese's Miniatures	0.279	81.86626
Twix	0.906	81.64291
Kit Kat	0.511	76.76860
Snickers	0.651	76.67378

Create a barplot to visualize the overall rankings

Q15: Make a first barplot of candy ranking based on winpercent values.

Answer Q15: Barplot provided below

```
# Create a gooplot barplot also known as geom_col
ggplot(candy) +
  aes(winpercent, rownames(candy)) +
  geom_col() +
  theme_bw()
```

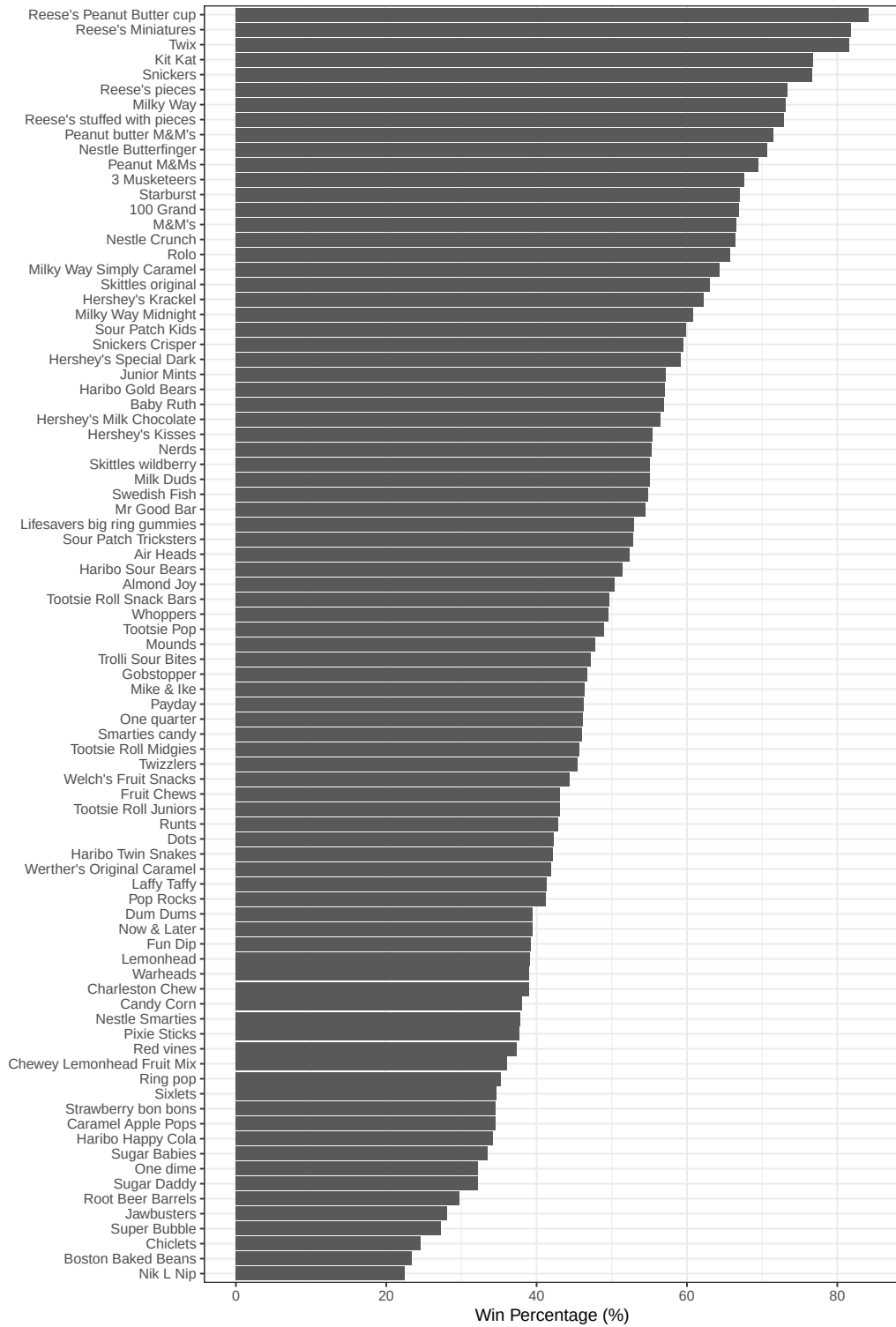


Q16: This is quite ugly, use the reorder() function to get the bars sorted by winpercent?

Answer Q16: I added `reorder()` function to sort the bar plot. I also added figure dimension and a title to make the graph look more similar to the one in the mini project. R code is shown below.

```
ggplot(candy) +  
  aes(winpercent, reorder(rownames(candy), winpercent)) +  
  geom_col() +  
  labs(x = "Win Percentage (%)", y = NULL, title = "Candy Rankings") +  
  theme_bw()
```

Candy Rankings



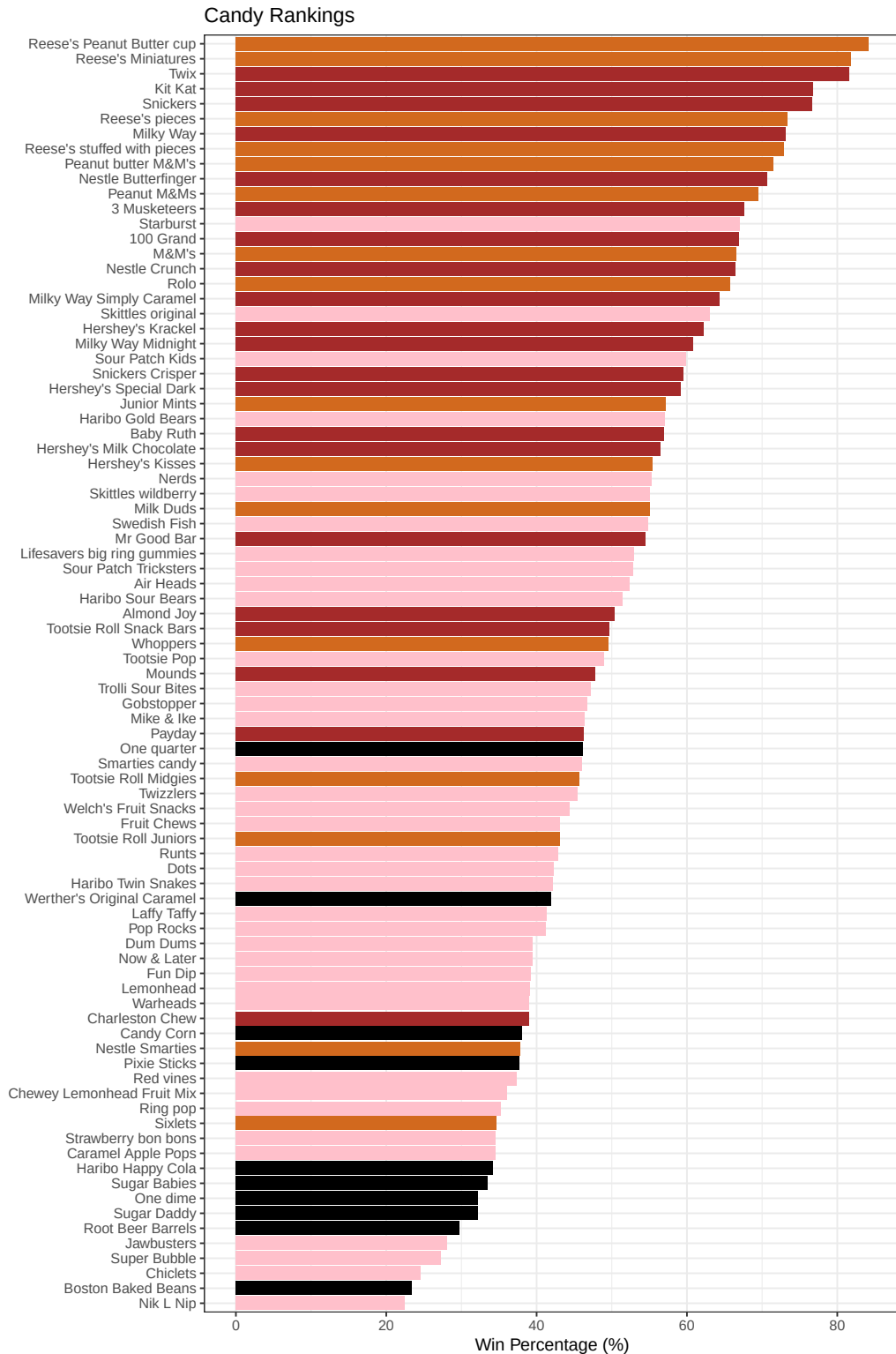
4.0.1 Time to add some useful color

Assign candy types to a color

```
# Setup a color vector (that signifies candy type)
my_cols=rep("black", nrow(candy))
my_cols[as.logical(candy$chocolate)] = "chocolate"
my_cols[as.logical(candy$bar)] = "brown"
my_cols[as.logical(candy$fruity)] = "pink"
```

Apply these color vectors to bar plot

```
# Adding color to barplot
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col(fill=my_cols) +
  labs(x = "Win Percentage (%)", y = NULL, title = "Candy Rankings") +
  theme_bw()
```



Q17: What is the worst ranked chocolate candy?

Answer Q17: Sixlets are the worst ranked chocolate candy with a winpercent of about 35% as estimated in the colored barplot. I found this by looking at the lowest orange bar on the plot because orange is the color assigned to chocolates.

Q18: What is the best ranked fruity candy?

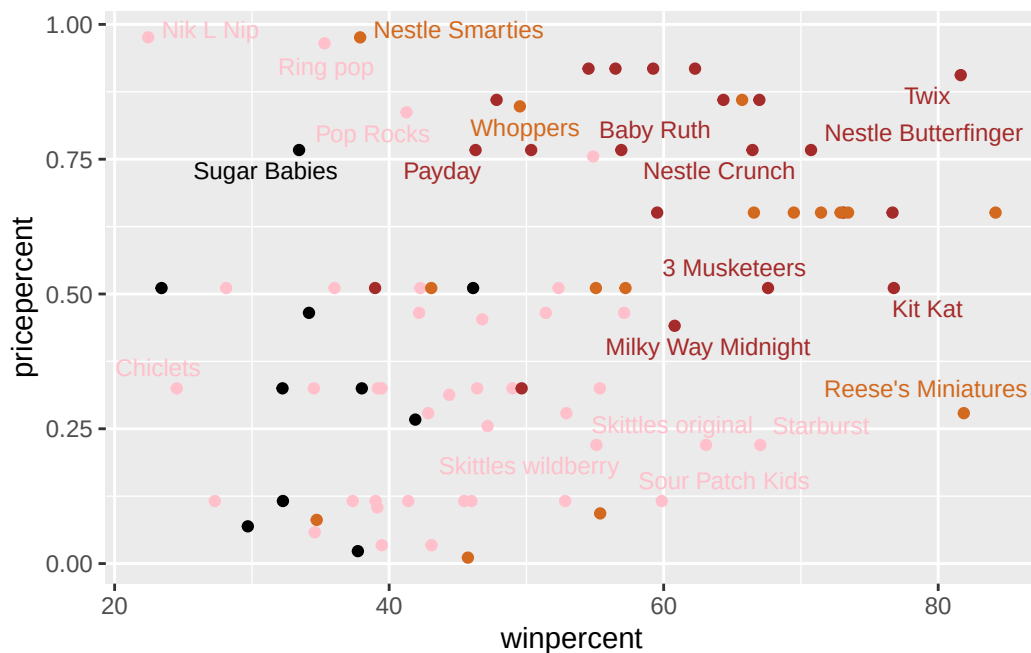
Answer Q18: Starbursts are the best ranked fruity candy with an estimated winpercent of 63%. Fruity candy was labeled as pink so I just looked for the pink bar plot that was ranked the highest.

Section 5: Taking a look at pricepercent

Make a plot of winpercent vs the pricepercent variable

```
library(ggrepel)

# How about a plot of win vs price
ggplot(candy) +
  aes(winpercent, pricepercent, label=rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 5)
```



Q19: Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

Answer Q19: In terms of highest winpercent and lowest pricepercent, the best candy would be “Reese’s Miniatures”; their estimated winpercent is about 80 and their estimated pricepercent is only about 0.26.

Q20: What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

Answer Q20: The top 5 most expensive candy types are: “Nik L Nip”, “Nestle Smarties”, “Ring pop”, “Hershey’s Krackel”, and “Hershey’s Milk Chocolate. Of these, the least popular is the “Nik L Nip” because it has the lowest winpercent being 22.45%. Table shown below.

```
# Sort by descending pricepercent
ord <- order(candy$pricepercent, decreasing = TRUE)
head( candy[ord,c(11,12)], n=5 )
```

	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

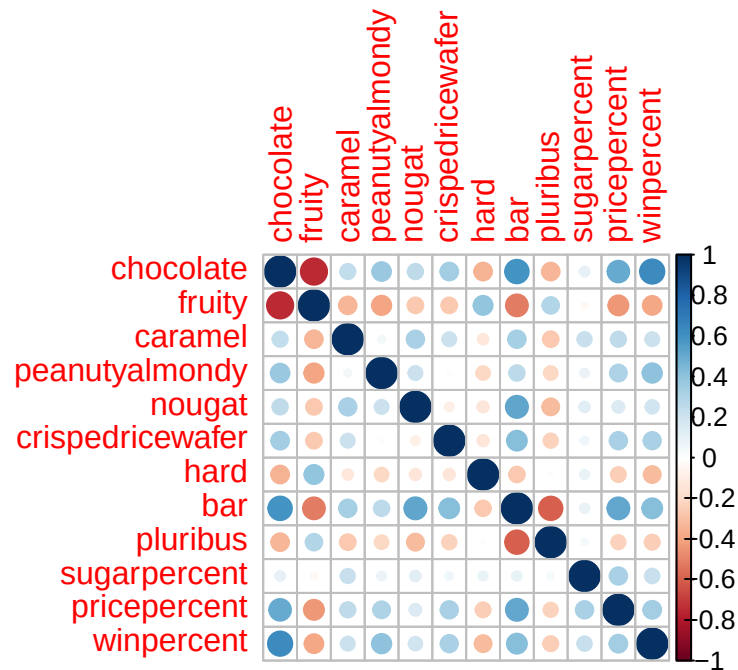
Section 6: Exploring the correlation structure

Plotting a Correlation Matrix

```
# Create a vector for candy correlations and Plot this
library(corrplot)
```

corrplot 0.95 loaded

```
cij <- cor(candy)
corrplot(cij)
```

Q22: Examining this plot what two variables are anti-correlated (i.e. have minus values)?

Answer Q22: The two variables that are anti-correlated are chocolate and fruit. The overlap of these two results in a large dark red circle which represents a large minus value and a negative correlation.

Q23: Similarly, what two variables are most positively correlated?

Answer Q23: Other than variables paired with themselves, the two variables that received the most positively correlated values were chocolate and bar and also chocolate and winpercent (these two seemed the most similar). This means that bar candies are more correlated with chocolate and also that chocolate is correlated with the higher winpercents.

Section7: Principal Component Analysis

Applying PCA with prcomp()

```
# Creating PCA
pca <- prcomp(candy, scale=TRUE)
summary(pca)
```

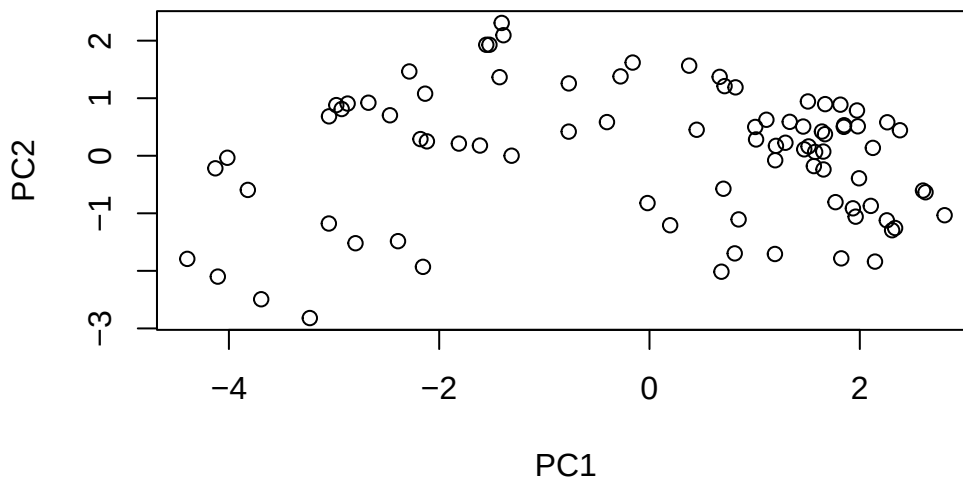
Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

	PC8	PC9	PC10	PC11	PC12
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000

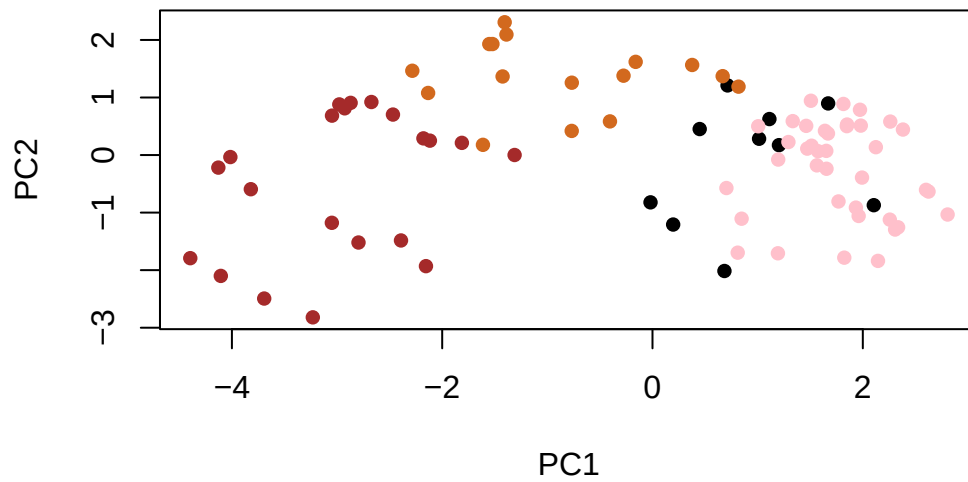
Plotting PCA

```
# PC1 vs. PC2 plot  
plot(pca$x[, 1:2])
```



Change Plot Character and Color

```
# Add Colors that were previously created for barplot  
plot(pca$x[, 1:2], col=my_cols, pch=16)
```

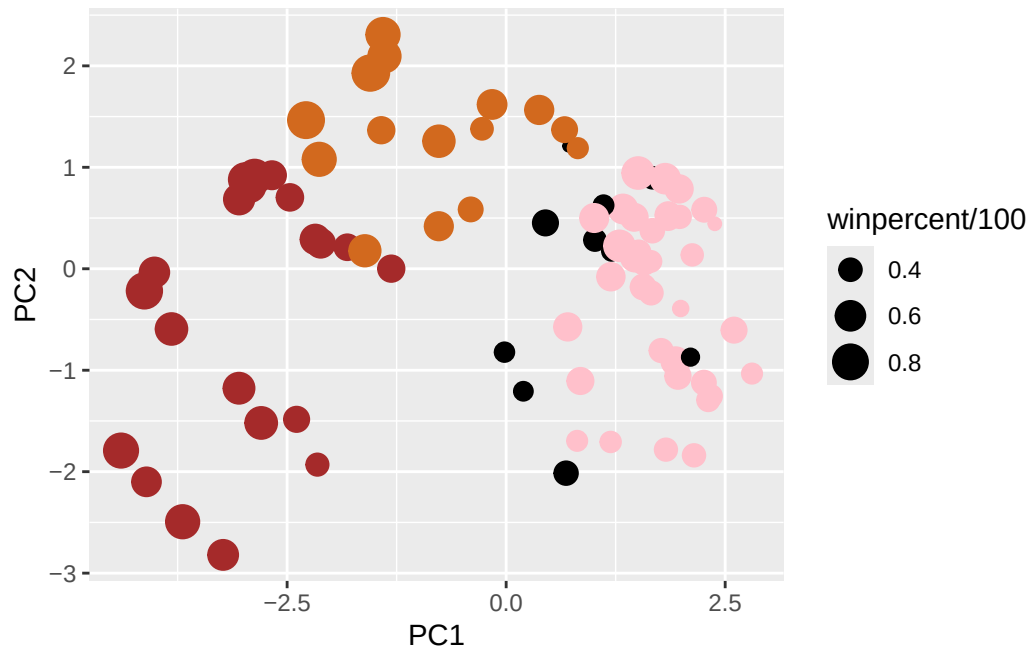


Create plot using ggplot2

```
# Make a new data-frame with our PCA results and candy data
my_data <- cbind(candy, pca$x[,1:3])

p <- ggplot(my_data) +
  aes(x = PC1, y = PC2,
      size=winpercent/100,
      text=rownames(my_data),
      label=rownames(my_data)) +
  geom_point(col=my_cols)

# View plot
p
```



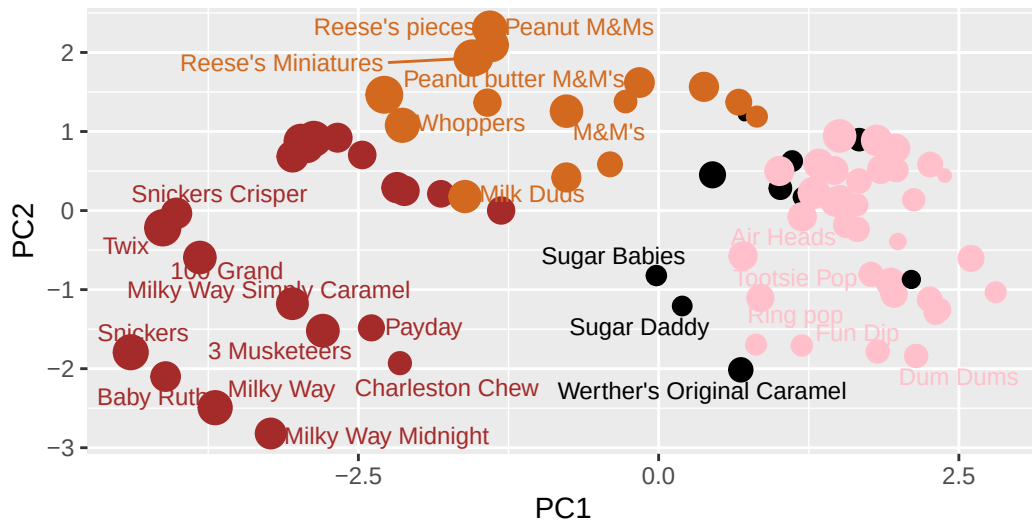
Plot using ggrepel

```
library(ggrepel)

p + geom_text_repel(size=3.3, col=my_cols, max.overlaps = 7) +
  theme(legend.position = "none") +
  labs(title="Halloween Candy PCA Space",
        subtitle="Colored by type: chocolate bar (dark brown), chocolate other (light brown),",
        caption="Data from 538")
```

Halloween Candy PCA Space

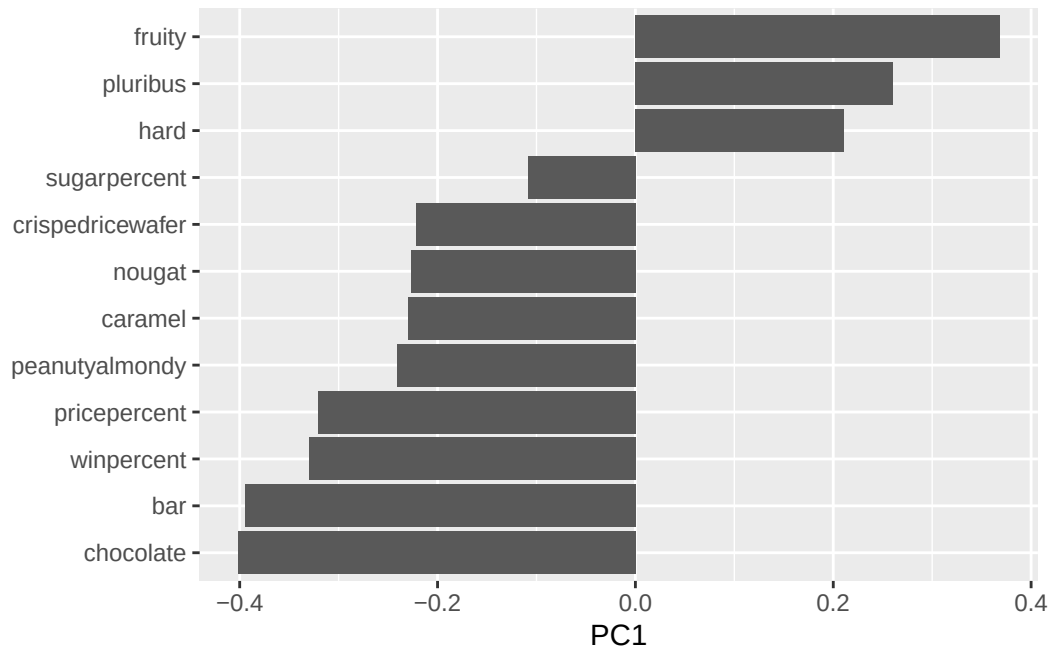
Colored by type: chocolate bar (dark brown), chocolate other (light brown),



Data from 538

PCA Loadings

```
# Generate Loading Plot
ggplot(pca$rotation) +
  aes(x=PC1, y=reorder(rownames(pca$rotation), PC1)) +
  geom_col() +
  labs(x = "PC1",
       y = NULL)
```



Q24: Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

- Answer Q24:** Loading Plot was created with the code provided *above*. The original variables that are picked up strongly by PC1 in the positive direction are: **“hard”**, **“pluribus”**, and **“fruity”**. Yes these do make sense to me because these three variables describe the typical fruit candy profiles. This suggests that PC1 is capturing the fundamental difference between fruity candies and chocolate-based candies. This relationship was previously seen in the corplot where fruity showed a strong positive correlation with pluribus and hard. We also saw that ruity showed a very strong negative (anti-correlation) with chocolate and bar which is represented by the strong positive direction in PC1.

8 Summary

Q25: Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

Answer Q25: Based on the PCA, the corrplot, and exploratory analysis the winning combination of characteristics are “**Chocolate**” and “**Bar**”. The exploratory analysis revealed the highest individual winpercents for chocolate and bars. The strongest correlations were also between these two. The PCA reveals the highest differences between fruit and chocolate but also revealed similarities between chocolate, bar, winpercent, and pricepercents. This cluster of characteristics reveals that the “winning” candy combination are chocolate bars.